M. Qasim Ayub

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Education

Lahore University of Management Sciences Lahore, Pakistan
B.Sc. Computer Science | Merit Scholarship Recipient **(Top 2% of class)** 3.98 CGPA
Relevant Coursework: Deep Vision-Language Models, Advanced Topics in Machine Learning, Foundations of AI and ML, Human-Computer Interaction, Agentic AI and LLMOps, Algorithms, Data Structures

International School Lahore Lahore, Pakistan
Grades: 4A* | 100% Merit Scholarship Recipient 2021 – 2023

Lahore Grammar School - JT Lahore, Pakistan
Grades: 9A* 1A | Distinction in Chemistry 2018 – 2021

Publications

M. Usman, **M. Q. Ayub**, U. Ahad, M. Tahir. **VisConf: Quality-Aware Consensus for Best-of-N in Vision-Language Model Reasoning** (arXiv, OpenReview). *NeurIPS 2026 Workshop on Grounded and Faithful Vision-Language Models for Real-World Deployment (VLM4RWD)*, Sydney. **Accepted**

Research

VisConf: Quality-Aware Test-Time Scaling for VLMs NeurIPS 2026 Workshop | 2026

- Developed **VisConf**, a rollout-quality score for Best-of-N inference that combines **visual groundedness**, measured from attention to image tokens, with **model confidence** measured by $D_{\mathrm{KL}}(U\|p)$ over the output distribution.
- Introduced **Rank-Weighted Consensus (RWC)**, which weights answer votes by rollout quality so that the final answer across up to **32 rollouts** is chosen by quality-aware voting rather than frequency alone. The method is verifier-free and training-free; across three VLMs on MathVista, MMMU-Pro and MMStar, it reached **56.36% mean accuracy** and outperformed all evaluated baselines in **24 of 27** settings.

Mechanistic Interpretability for Post-Translational Modification Prediction In progress

- Reproducing **SAPP** and **GraphPhos** as baselines and building a pipeline that predicts PTM sites from protein sequence and residue-level structure, replacing summaries based only on relative solvent accessibility (RSA) with contact graphs and confidence-aware geometry.
- Investigating the **mechanistic basis** of why these methods perform well, in order to identify interpretable architectural signals and design more accurate and robust PTM predictors.

Open-Set Recognition in Federated Learning In progress

- Developed latent-space augmentations that generate **synthetic negatives**, and a KL-divergence regularizer that encourages them to have equal affinity to all class prototypes, improving separation from known classes.
- Formulating a **discriminative-subspace hypothesis** to explain why particular augmentations improve open-set recognition and which representation directions matter most under federated heterogeneity.

Enviromon: Game-Based Environmental Learning (Live) Human-Computer Interaction | 2026

- Designed a **bilingual (English and Urdu)** educational game for children in Pakistan in which nine missions map environmental issues, such as smog, waste sorting and industrial leaks, onto familiar arcade mechanics; each mission begins by explaining the issue.

Health × AI

Mederium (mederium.com) Lahore, Pakistan | 2026 – Present

- Subsidized health-technology platform that provides partner clinics with digital infrastructure and serves as a setting to study **what improves patient satisfaction** and how doctors can collect **structured, research-ready clinical data**.
- Built voice-first documentation in Urdu and English that drafts **ICD-10-coded** records for doctor review, a patient assistant grounded in each clinic's knowledge base, and a doctor copilot that acts only after explicit confirmation.

Experience

Teaching Assistant - Advanced Topics in Machine Learning | LUMS 2026 – Present

- Conduct recitations and office hours, and help develop and grade **assignments, quizzes and technical assessments**.

GenAI Engineer | Technovate Global (Company Profile) Remote | March 2026 – October 2026

- Developed **multi-stage agentic workflows** with **self-validation loops**, and RAG pipelines covering **ingestion, metadata pre-filtering, hybrid retrieval and reranking**, supported by 130+ tests for security, correctness and concurrency.

Teaching Assistant - Deep Learning | MS AI LUMS January 2026 – June 2026

- Designed assignments on **advanced machine learning algorithms** and **model interpretability**, led recitations, and built a **live leaderboard portal** that automated grading for **100+ students**.

Teaching Assistant - Coding for Careers | LUMS August – December 2025

- Evaluated **30+ technical interviews**, provided targeted feedback, and held bi-weekly office hours on coding assignments.

Software Engineer and QA | DevClan (Company Profile) Hybrid | June 2025 – October 2025

- Built a Power BI dashboard covering **100,000+ properties** for a US real-estate agency and automated workflows, reducing manual work by up to **90%**; developed and tested CRM components for **LendingPad**.

Leadership

Convenor | PsiFi, SPADES 2025 – 2026

- Led the largest edition of Pakistan's largest science event, hosting **1,500+ delegates** from every province, with a focus on broadening students' access to opportunities that remain limited in Pakistan.
- Launched the **Psynergy Fair** to connect students with academia and industry, and introduced, in partnership with Pakistan's **IOAA** team, a **PKR 50,000 scholarship** for each of the top 10% of delegates, covering registration and 3–4 months of coaching.

Course Research

Advanced Topics in Machine Learning: Model Compression (Manuscript), Domain Adaptation and Generalization (Manuscript), Federated Learning for Heterogeneous Data (Manuscript), LLM Decoding, Alignment and Interpretability (Manuscript)

Deep Vision-Language Models: SFT, PPO, DPO, GRPO and RLVR post-training; DDPM from scratch with ablations and a DDIM extension

Skills and Proficiency

Languages: Python, C, C++, JavaScript

Research: PyTorch, Vision-Language Models, Test-Time Scaling, Mechanistic Interpretability, Protein Language Models, Graph Neural Networks, Federated Learning, Reinforcement Learning, Diffusion Models, Model Compression

Engineering: RAG, LangGraph, LangChain, FastAPI, PostgreSQL, Qdrant, Git